

HEMANTH RAJ M V

Bengaluru, Karnataka

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Education

BMS College of Engineering, Bengaluru
Bachelor of Engineering in Computer Science (Data Science)

Oct 2023 – Present
CGPA: 8.32

Technical Skills

Languages: Python, SQL, Kotlin, Java, C

ML & Data: Pandas, NumPy, Scikit-learn, TensorFlow, Feature Engineering, NLP (NLTK), EDA

Big Data & MLOps: PySpark, Apache Spark, Spark MLlib, FastAPI, REST APIs, Docker, RAG Pipelines

Databases & Tools: ChromaDB, PostgreSQL, MongoDB, Power BI, Tableau, Git, Jupyter Notebook

Projects

- **Tether — Social Accountability Android App** GitHub — APK (v1.0)
Kotlin, MVVM, Firebase Auth/Firestore/FCM, Navigation Component
 - Built a production-grade Android app using **Kotlin/MVVM**; optimized navigation with manual **NavController** handling to eliminate UI flickering and achieve **sub-second tab switching**.
 - Engineered a **Foreground Service** for multi-mode focus tracking (Stopwatch/Pomodoro) with **1-second polling accuracy**, ensuring **100% session persistence** under battery-restricted environments.
 - Implemented a **real-time nudge system** using **FCM + Firestore** with **sub-second sync**; optimized backend via **composite indexing** to efficiently handle thousands of log entries by group and date.
- **Visual Document Q&A — End-to-End RAG Pipeline** GitHub
FastAPI, ChromaDB, Ollama (phi3), PyMuPDF
 - Built a **Visual Document Q&A system** with a **FastAPI + ChromaDB** RAG pipeline using **all-MiniLM-L6-v2** embeddings and **Ollama (phi3)**, achieving **~12 tok/s** streaming via SSE with **1s retrieval latency**.
 - Engineered a **4-signal Concept Importance Engine** (TF-IDF + spread + heading weight + **PageRank centrality**) and visualized concept relationships as an **Obsidian-style D3.js force-directed graph** with tiered node coloring, co-occurrence edges, and click-to-query interaction.
 - Implemented **tiered heading detection** (H1/H2/H3 via PyMuPDF font-size analysis) and **async LLM topic refinement** that renders the TF-IDF graph instantly then merges domain concepts in background — boosting validated terms by **1.5×** with silent fallback ensuring zero UX disruption.
- **YouTube Global Trends Analytics** GitHub Live Demo
PySpark, Spark MLlib, Streamlit, Plotly, Pandas
 - Engineered an end-to-end **Big Data pipeline** using **PySpark** to process **5.5GB / 10M+ records** across **113 countries**.
 - Trained a **RandomForestRegressor** (50 trees, depth 10) via **Spark MLlib** on 8 engineered features with strict **train-test split before aggregate computation** to prevent data leakage, achieving **$R^2 = 0.976$** on held-out test data.
 - Deployed an interactive **Streamlit + Plotly dashboard** with region/language filtering, geographic performance mapping, and engagement scatter plots — optimized via stratified sampling for **sub-second response times**.

Achievements / Leadership

Nutanix Certified Associate (NCA) 2026

- Validated core competency in **hyperconverged infrastructure (HCI)** and **cloud infrastructure fundamentals**.

GenAI Hackathon — Special Mention 2025

- Recognized for **Visual Document Q&A**, a RAG-based system with FastAPI, ChromaDB, and a D3.js visualization layer.

Treasurer, DSync Club — BMSCE

April 2025 – Present

- Managed end-to-end club finances across **10+ events**, owning budgeting, expense tracking, and reimbursements. Secured corporate sponsorships for **UTSAV** (60K+ footfall) and **PHASESHIFT** (40K+ footfall)